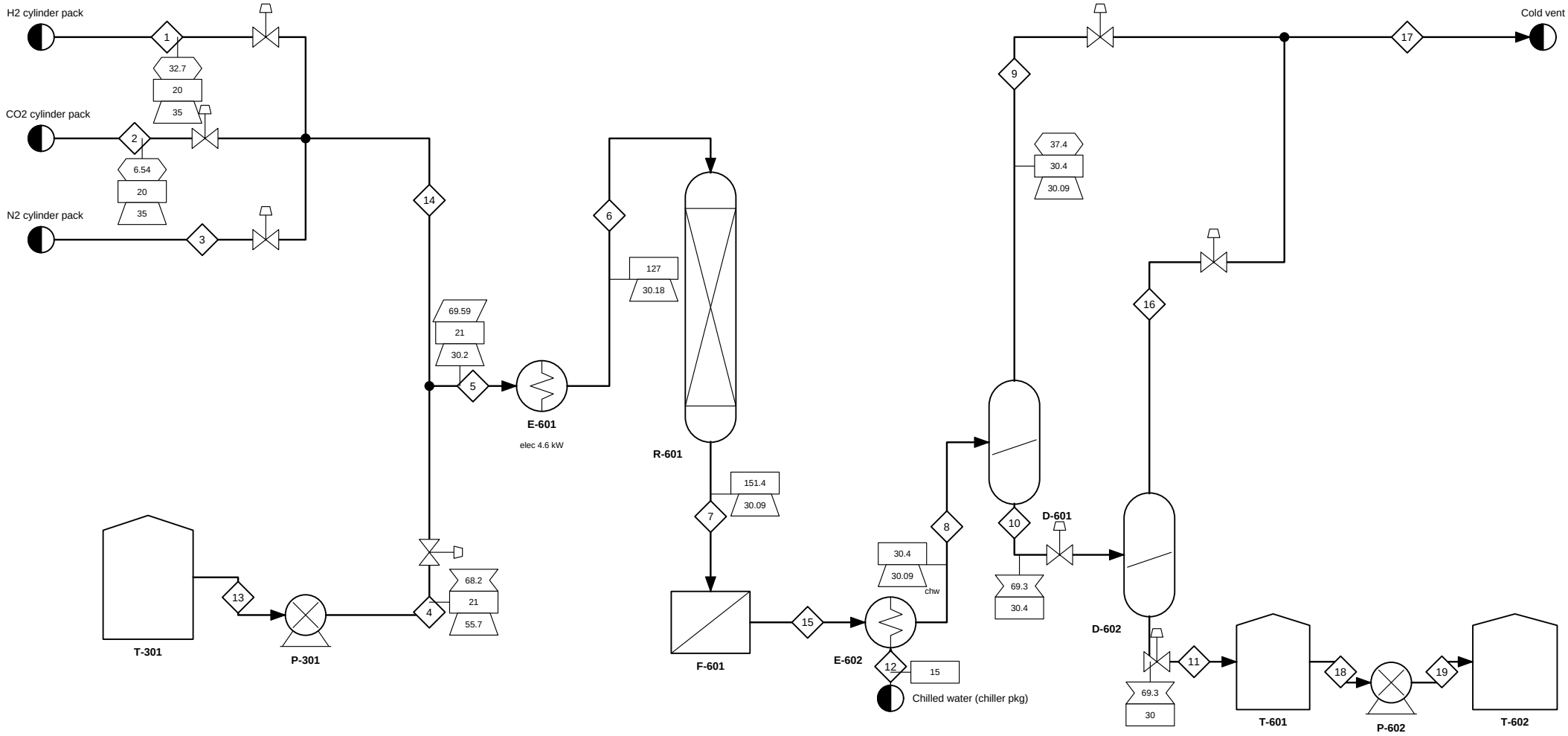
T-301	P-301	E-601	R-601	F-601	E-602	D-601	D-602	T-601	P-602	T-602
Ethanol Storage Tank	Ethanol Metering Pump	Electric Feed Preheater	Trickle-Bed Reactor	Outlet Filter	Product Condenser	HP Separator	LP Flash Drum	Methanol Product Tank	Transfer Pump (air)	Waste Tank (IBC)



REACTIONS (Cu/ZnO/Al2O3, 30 bar g, ~150 degC)

CO2 + 3H2 <-> MeOH + H2O (direct)

CO2 + H2 + EtOH -> EF + H2O; EF + 2H2 -> MeOH + EtOH

CO2 + H2 <-> CO + H2O; 2EtOH <-> EA + 2H2

NOTES

1. Redrawn from 2407-PE-PFD-001 R1 and HMB 2407-PE-HMB-001 R1. Flags: OTS design point (Case 3, LHSV 0.30 1/h), model values.
2. Streams 1-12 = Process-tab / BFD numbers; 13-19 drawing only. Gas NL/min, liquid mL/min, mass g/min, T degC, P bar g.
3. R-601 band heaters E1/E2/E3 (TIC-611/612/613) assumed tags. Cylinder side of packs: H2 150, CO2 68 bar g (HMB).

LEGEND

Gas flow (NL/min) · Liquid flow (mL/min) · Mass flow (g/min) · Temperature (degC) · Pressure (bar g) · Process Line

Utility Line · Control Valve · Process Input / Output · Stream Number · Line Crossing · Storage Tank

PD Pump · Heat Exchanger · Packed-Bed Reactor · Filter · Separator

MTJA CO2-METHANOL PILOT · DEPT. OF CHEMICAL ENGINEERING, CHULALONGKORN UNIVERSITY

PROCESS FLOW DIAGRAM

PILOT TRICKLE-BED REACTOR R-601 · CO2 + H2 TO METHANOL (ETHANOL-ASSISTED)

DRAWING No.	REV.	SIZE	SHEET
R601-PFD-001	0	A4	1 OF 1
DRAWN	**CHECKED**	**SCALE**	**DATE**
Apinan Soottitantawat		NTS	2026-10-24